

Chapter 115

Position Nodes

This chapter details the Position nodes available in Fusion.

The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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Volume Fog [VLF]



The Volume Fog node

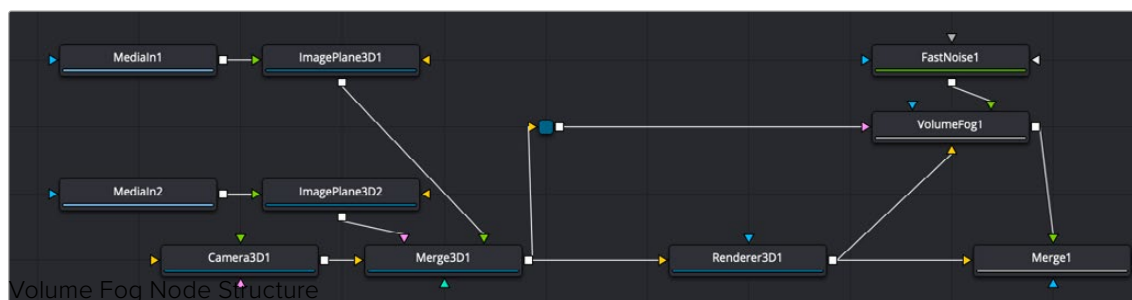
Volume Fog Node Introduction

The Volume Fog node is used to create sophisticated volumetric fog on images containing XYZ Position channels.

As opposed to 3D-rendered volumetric fog, it works on 2D images and delivers much faster results and interactive feedback when setting up the fog. See the “WPP Concept” section at the end of this chapter for further explanation of how this technology works and to learn about the required imagery.

Basic Node Setup

The Volume Fog node takes an image input; in the example below, it is a Renderer 3D with World Position enabled in the output channels. Another input is the 3D scene, which contains the camera. A Fast Noise node generates the fog texture.



A 3D scene and a rendered scene connected to Volume Fog

Inputs

The following inputs appear on the Volume Fog node in the Node Editor.

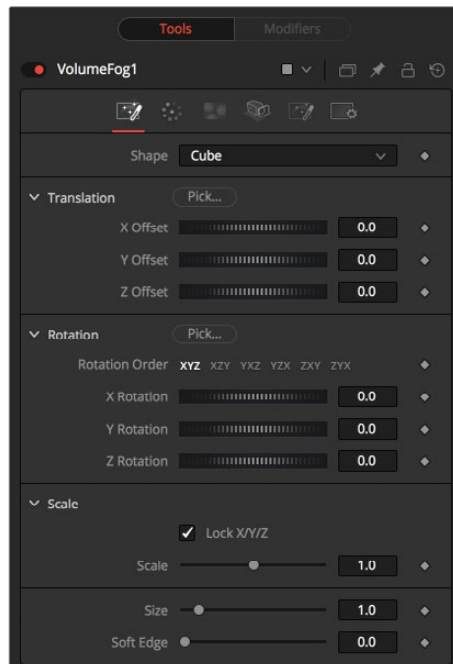
Image: The orange input accepts the primary image where the fog will be applied. This image contains a World Position Pass in the XYZ Position channels.

Fog Image: The green Fog image input is for creating volumetric fog with varying depth and extent; a 2D image can be connected here. A good starting point is to use a Fast Noise at a small resolution of 256 x 256 pixels.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the fog to certain areas.

Scene Input: The magenta scene input accepts a 3D scene containing a 3D Camera.

Inspector



The Volume Fog Shape tab

Shape Tab

The Shape tab defines the size and location of the fog volume. You can either use the Pick buttons to select the location and orientation in the viewer or use the Translation, Rotation, and Scale controls.

Shape

This menu switches between a basic spherical or rectangular volume to be placed in your image. These volumes can then be further refined using the Fog image and effect mask.

Pick

Drag the Pick button into the viewer to select the XYZ coordinates from any 3D scene or 2D image containing XYZ values, such as a rendered World Pass, to position the center of the Volume object. When picking from a 2D image, make sure it's rendered in 32-bit float to get full precision.

X, Y, Z Offset

These controls can be used to position the center of the fog volume manually or can be animated or connected to other controls in Fusion.

Rotation Pick

Drag the Pick button into the viewer to select the rotational values from any 3D Scene or 2D image containing those values, like an XYZ-Normal-Pass, to reorient the fog volume.

When picking from a 2D image, like an XYZ Normal pass, make sure it's rendered in 32-bit float to get full precision and accurate rotational values.

X, Y, Z Rotation

Use these controls to rotate the fog volume around its center.

X, Y, Z Scale

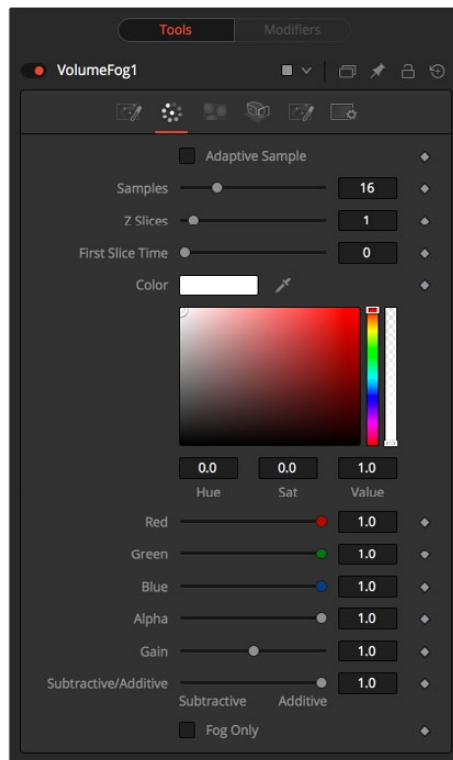
Scale the fog volume in any direction from its center to refine further the overall Size value specified below.

Size

The overall size of the fog volume created.

Soft Edge

Controls how much the fog volume is faded toward the center from its perimeter to achieve a softer look.



The Volume Fog Color tab

Color Tab

The Color tab controls the detail and color of the fog.

Adaptive Samples

Volumes images consist of multiple layers, so there may be 64 layers in a volume. This checkbox adjusts the rendering algorithm for how to best blend those layers.

Dither: Applies a form of noise to improve the blending and hide visible layer differences.

Samples

Determines how many times a “ray” shot into the volume will be evaluated before the final image is created. Not unlike raytracing, higher values lead to more detail inside the volume but also increase render times.

Z Slices

The higher the Z Slices value, the more images from the connected Fog image sequence will be used to form the depth of the volume.

You can, for example, use a Fast Noise with a high Seethe Rate to create such a sequence of images. Be careful with the resolution of the images. Higher resolutions can require a large amount of memory. As a rule of thumb, a resolution of 256 x 256 pixels with 256 Z Slices (i.e., forming a 256 x 256 x 256 cubic volume, which will use up to 256 MB for full color 32-bit float data) should give you a good starting point.

First Slice Time

Determines which frame of the Global Range is used to deliver the first slice from the connected fog image sequence.

Make sure that both Global In and Global Out, as well as the valid range of your source node, fall within the range of First Slice Time + Z Slices.

Color

Allows you to modify the color of the fog generated. This will multiply over any color provided by the connected Fog image.

Gain

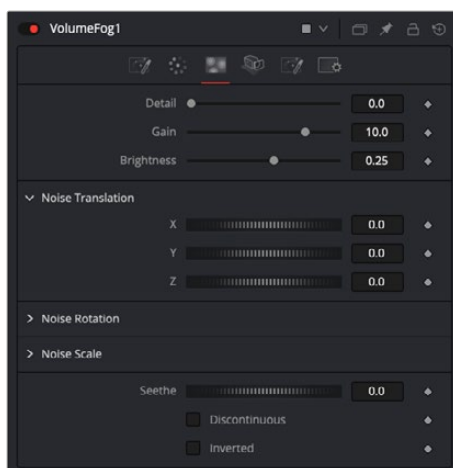
Increases or decreases the intensity of the fog. More Gain will lead to a stronger glow and less transparency in the fog. Lower values let the fog appear less dense.

Subtractive/Additive Slider

Similar to the Merge node, this value controls whether the fog is composed onto the image in Additive or Subtractive mode, leading to a brighter or dimmer appearance of the fog.

Fog Only

This option outputs the generated fog on a black background, which then can be composited manually or used as a mask on a Color Corrector for further refinement.



The Volume Fog Noise tab

Noise Tab

The Noise tab controls the shape and pattern of the noise added to the fog.

Detail

Increase the value of this slider to produce a greater level of detail in the noise result. Larger values add more layers of increasingly detailed noise without affecting the overall pattern. High values take longer to render but can produce a more natural result.

Gain

This control increases or decreases the brightest parts of the noise map.

Brightness

This control adjusts the overall brightness of the noise map, before any gradient color mapping is applied. In Gradient mode, this produces a similar effect to the Offset control.

Translation

Use the Translation coordinate control to pan and move the noise pattern.

Noise Rotation

Use the Rotation controls to orient the noise pattern in 3D.

Seethe

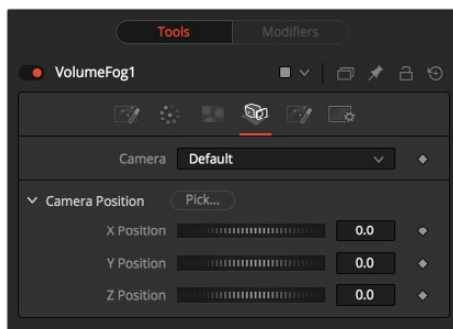
Adjust this thumbwheel control to interpolate the noise map against a different noise map. This will cause a crawling shift in the noise, like it was drifting or flowing. This control must be animated to affect the noise over time.

Discontinuous

Normally, the Noise function interpolates between values to create a smooth, continuous gradient of results. Enable this checkbox to create hard discontinuity lines along some of the noise contours. The result will be a dramatically different effect.

Inverted

Select this checkbox to invert the noise, creating a negative image of the original pattern. This is most effective when Discontinuous is also enabled.



The Volume Fog Camera tab

Camera Tab

For a perfect evaluation of a fog volume, a camera or 3D scene can be connected to the Scene input of the node.

Camera

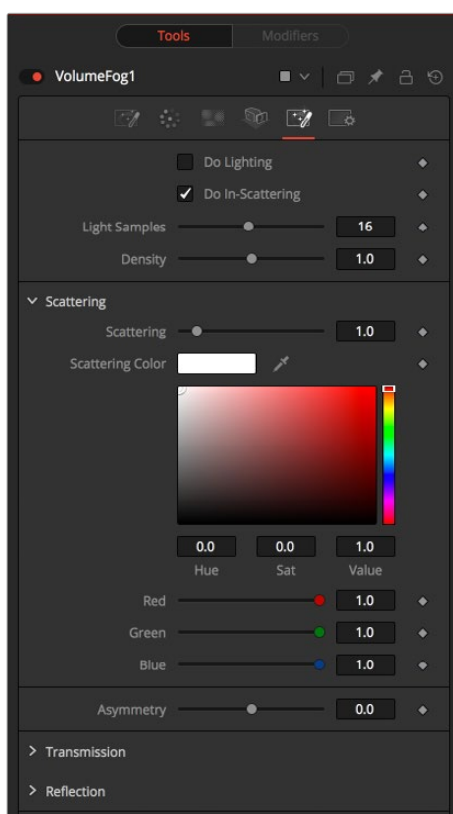
If multiple cameras are available in the connected Scene input, this menu allows the selection of the correct camera needed to evaluate the fog volume. Instead of connecting a camera, position values can be provided manually or by connecting the XYZ values to other controls.

Translation Pick

Drag the Pick button into the viewer to select XYZ coordinates from any 3D scene or 2D image containing XYZ values, like a rendered World Pass, to define the center of the camera. When picking from a 2D image, make sure it's rendered in 32-bit float to get full precision.

X, Y, Z Offset

These controls can be used to define the center of the camera manually or can be animated or connected to other controls in Fusion.



The Volume Fog Light tab

Light Tab

To utilize the controls in the Light tab, you must have actual lights in your 3D scene. Connect that scene, including Camera and Lights, to the 3D input of the node.

Do Lighting

Enables or disables lighting calculations. Keep in mind that when not using OpenCL (i.e., rendering on the CPU), these calculations may become a bit slow.

Do In-Scattering

Enables or disables light-scattering calculations. The volume will still be lit according to the state of the Do Lighting checkbox, but scattering will not be performed.

Light Samples

Determines how accurate the lighting is calculated. Higher values mean more accurate calculation at the expense of longer render times.

Density

This is similar to scattering in that it makes the fog appear thicker. With a high amount of scattering, though, the light will be scattered out of the volume before it has had much chance to travel through the fog, meaning it won't pick up a lot of the transmission color. With a high density instead, the fog still appears thicker, but the light gets a chance to be transmitted, thus picking up the transmission color before it gets scattered out. Scattering is affected by the light direction when Asymmetry is not 0.0. Density is not affected by light direction at all.

Scattering

Determines how much of the light bouncing around in the volume ends up scattering the light out of the fog. If the light scatters more, or more accurately, then there's a higher probability of the light being scattered out of the volume, hence less light is left to continue through the fog. This option can make the fog seem denser.

Asymmetry

Determines in what direction the light is scattered. A value of 0 produces uniform, or isotropic, scattering, meaning all directions have equal probability. A value greater than 0 causes "forward scattering," meaning the light is scattered more into the direction of the light rays. This is similar to what happens with water droplets in clouds. A value smaller than 0 produces "back scattering," where the light is more scattered back toward the original light source.

Transmission

Defines the color that is transmitted through the fog. The light that doesn't get scattered out will tend toward this color. It is a multiplier, though, so if you have a red light, but blue transmission, you won't see any blue.

Reflection

Changes the intensity of the light that is scattered out. Reflection can be used to modify the overall color before Emission is added. This will be combined with the color channels of the volume texture and then used to scale the values. The color options and the color channels of the volume texture are multiplied together, so if the volume texture were red, setting the Reflection color options to blue would not make the result blue. In such a case, they will multiply together to produce black.

Emission

This adds a bit of "glowing" to the fog, adding energy/light back into the calculation. If there are no lights in the scene, and the fog emission is set to be 1.0, the results are similar to no lighting, like turning off the Do Lighting option. Glowing can also be done while producing a different kind of look, by having a Transmission greater than 1. This, however, would never happen in the real world.

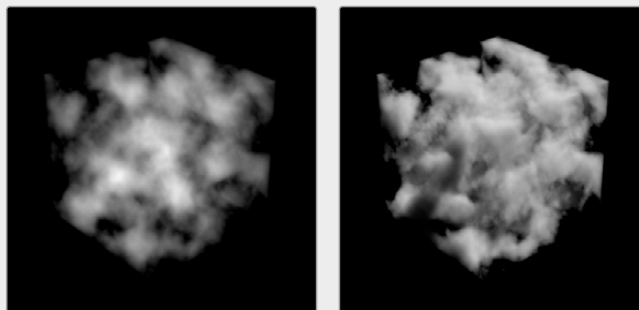
Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Position nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Examples

In these examples, we are looking at a volume from the outside. On the left, you see how the Volume Fog looks with straight accumulation. That means the Do Lighting option is turned off.



On the right, you see the same volume with lighting/scattering turned on, and a single point light.

Here, we have a slightly more complex Volume.



On the left with straight accumulation; in the middle with lighting, scattering, and a single point light; and on the right, the light in the scene has been moved, which also influences the look of the volume.

Volume Mask [VLM]



The Volume Mask node

Volume Mask Node Introduction

The Volume Mask node is used to create volumetric masks from images containing XYZ Position channels.

This can, for example, be used to isolate objects for color correction without the need to track or rotoscope the scene. See the “WPP Concept” section later in this chapter for further explanation on how this technology works and to learn about the required imagery.

Inputs

The following three inputs appear on the Volume Mask node in the Node Editor:

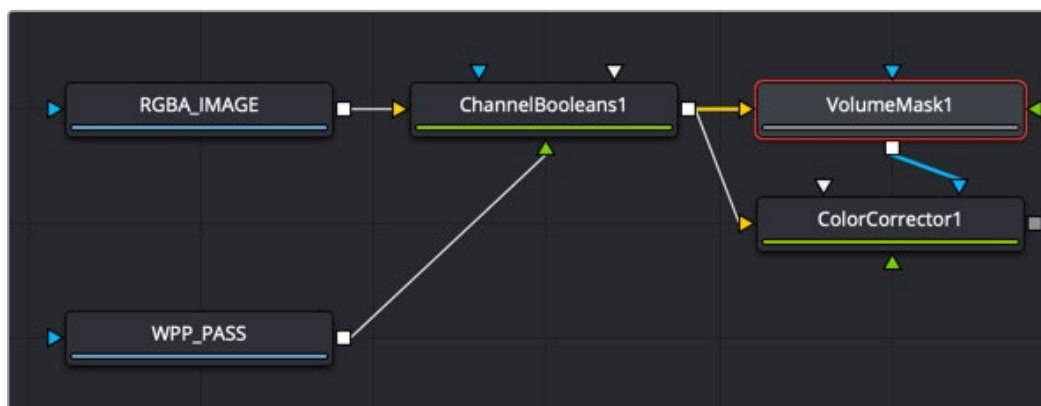
Image: The orange image input accepts a 2D image containing a World Position Pass in the XYZ Position channels.

Mask Image: An image can be connected to the green mask image input for refining the mask.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the volume mask to certain areas.

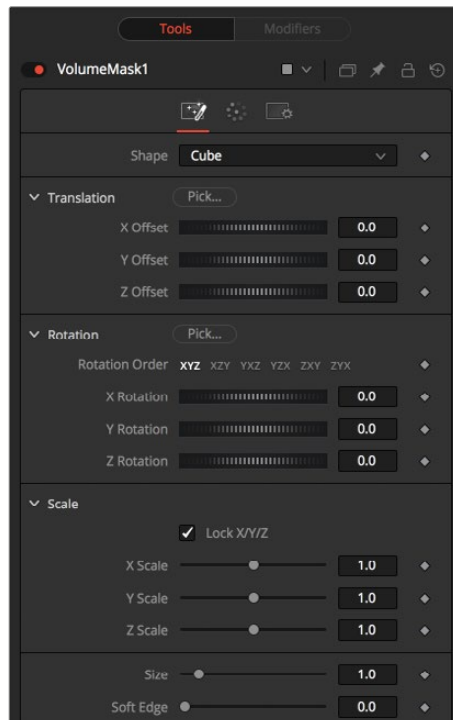
Basic Node Setup

Below, a MediaIn labeled RGBA contains the main rendered image from a 3D scene. A World Position Pass from a 3D-rendered scene is labeled WPP_PASS. The Channel Booleans node is used to map the X position pass to the red channel, Y to the green channel, and Z to the blue channel. The Volume Mask tool extracts the 3D position information from the World Position Pass to place a mask in 3D space. The mask is then used as an effect mask on a color corrector to selectively color correct objects in a 3D scene.



A Volume Mask tool takes advantage of World Position Pass for color correction in a 3D scene

Inspector



The Volume Mask Shape tab

Shape Tab

The Shape tab defines the size and location of the Volume Mask. You can either use the Pick buttons to select the location and orientation in the viewer or use the Translation, Rotation, and Scale controls.

Shape

This menu switches between a spherical or rectangular mask to be placed in your image. The mask can be further refined using the mask image input.

Translation Pick

Drag the Pick button into the viewer to select XYZ coordinates from any 3D scene or 2D image containing XYZ values, like a rendered World Pass, to position the center of the Volume Mask. When picking from a 2D image, make sure it's rendered in 32-bit float to get full precision.

X, Y, Z Offset

These controls can be used to position the center of the mask manually or can be animated or connected to other controls in Fusion.

Rotation Pick

Drag the Pick button into the viewer to select rotational values from any 3D scene or 2D image containing those values, like an XYZ Normal pass, to reorient the mask.

When picking from a 2D image, like an XYZ Normal pass, make sure it's rendered in 32-bit float, and use World Space coordinates to get full precision and the correct rotational values.

X, Y, Z Rotation

Use these controls to rotate the mask around its center.

X, Y, Z Scale

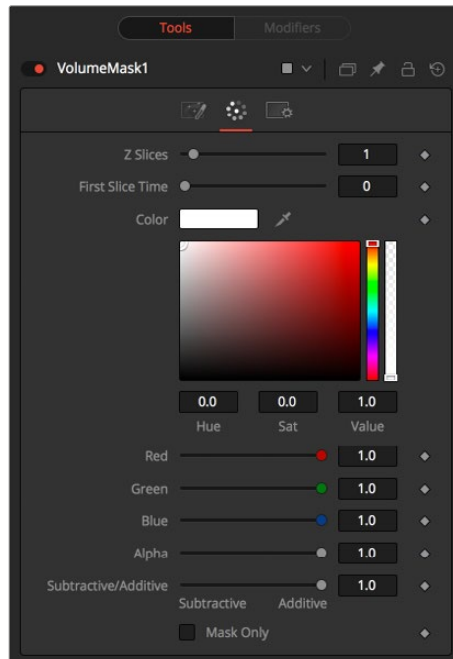
Scale the mask in any direction from its center to further refine the overall Size value specified below.

Size

The overall size, in X, Y, and Z, of the mask created.

Soft Edge

Controls how much the Volume is faded toward the center from its perimeter to achieve a softer look.



The Volume Mask Color tab

Color Tab

The Color tab controls the color and blending of the mask image.

Color

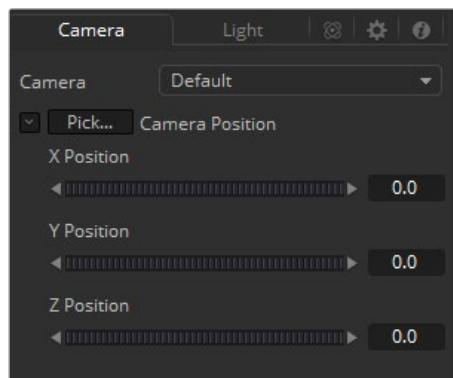
Allows you to modify the color of the generated Volume Mask. This will add to any color provided by the connected mask image.

Subtractive/Additive Slider

Similar to the Merge node, this value controls whether the mask is composed onto the image in Additive or Subtractive mode, leading to a brighter or dimmer appearance of the mask.

Mask Only

Outputs the generated mask on a black background, which then can be used as a mask on a Color Corrector for further refinement.



The Volume Mask Camera tab

Camera Tab

For a perfect evaluation of a Volume, a camera or 3D scene can be connected to the Scene input of the node.

Camera

If multiple cameras are available in the connected Scene input, this drop-down menu allows you to choose the correct camera needed to evaluate the Volume.

Instead of connecting a camera, position values can also be provided manually or by connecting the XYZ values to other controls.

Translation Pick

Drag the Pick button into the viewer to select XYZ coordinates from any 3D scene or 2D image containing XYZ values, like a rendered World Pass, to define the center of the camera.

When picking from a 2D image, make sure it's rendered in 32-bit float to get full precision.

X, Y, Z Offset

These controls can be used to define the center of the camera manually or can be animated or connected to other controls in Fusion.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Position nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Z to World Pos [Z2W]



The Z to World Position node

Z to World Pos Node Introduction

The Z to World Pos node is used to either generate a World Position Pass from a Z channel and a 3D Camera or a Z channel from a World Position Pass and a 3D Camera.

Creating a World Position Pass from Z-depth can be useful when your 3D application is not capable of creating a WPP.

It can also be used when a 3D-tracking software outputs a per-pixel Z-depth together with the 3D Camera. Thus, the Volume Mask and Volume Fog could be applied to real-world scenes. The quality of the resulting WPP depends mainly on the quality of the incoming Z channel.

See the “WPP Concept” section for further explanation on how this technology works and to learn about the required imagery.

Inputs

The following inputs appear on the node tile in the Node Editor:

Image: The orange image input accepts an image containing a World Position Pass or a Z-depth pass, depending on the desired operation.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the World Position Pass to certain areas.

Scene Input: The magenta scene input accepts a 3D scene input containing a 3D Camera.

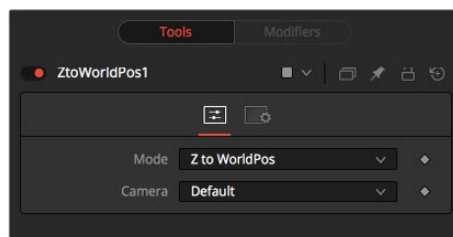
Basic Node Setup

Below, a MediaIn labeled RGBA contains the main rendered image from a 3D scene. A Z-depth pass from a 3D-rendered scene is labeled Z_PASS. The Channel Booleans node is used to map the Aux Z channel into either the red, green, or blue channel. The Z to World Position node is placed after the Channel Booleans node, and an imported 3D camera that matches the RGBA image is connected to the 3D camera input on the Z to World Position node. A Channel Booleans node is placed after the Z to World Position node, which can remap the X, Y, and Z positions for use in other nodes.



A Z to World Position node creates a World Position Pass from a Z-depth pass

Inspector



The Z to World Position Controls tab

Controls Tab

The Controls tab determines whether you are creating a World Position Pass or a Z channel. If there is more than one camera in the connected scene, this tab also selects the camera to use for the calculation.

Mode

This menu switches between creating a Z channel from a World Position Pass or vice versa.

Camera

If multiple cameras are available in the connected Scene input, this drop-down menu allows you to choose the correct camera needed to evaluate the image.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Position nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

WPP Concept

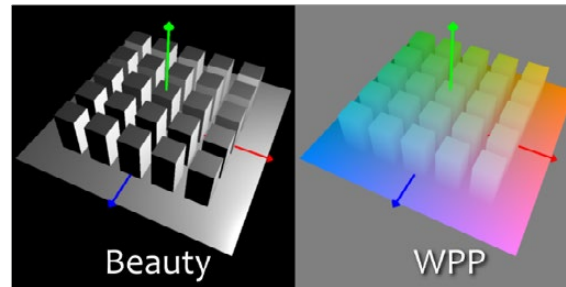
The Position nodes in Fusion offer an entirely new way of working with masks and Volumetrics for footage containing XYZ Position channels. Z to World offers the option to create those channels out of a Z channel and 3D Camera information. For this overview, we refer to the World Position Pass as WPP.

What Is a WPP?

The WPP interprets each pixel’s XYZ position in 3D space as an RGB color value.

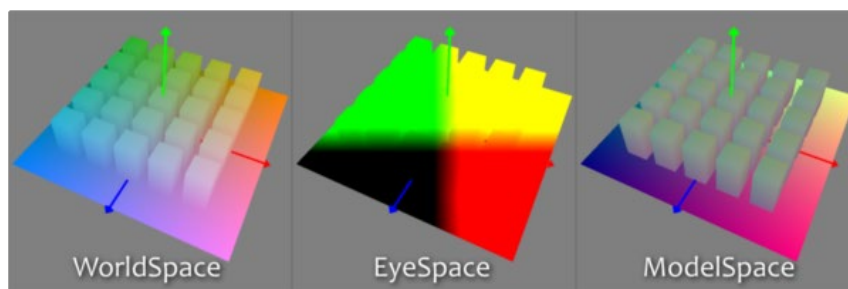
For instance, if a pixel sits at 0/0/0, the resulting pixel has an RGB value of 0/0/0 and thus will be black. If the pixel sits at 1/0/0 in the 3D scene, the resulting pixel is entirely red. Of course, if the coordinates of the pixel are something like -60/75/123, WPP interprets those values as RGB color values as well.

Due to the potentially enormous size of a 3D scene, the WPP channel should always be rendered in 32-bit floating point to provide the accuracy needed. The image below shows a 3D rendering of a scene with its center sitting at 0/0/0 in 3D Space and the related WPP. For better visibility, the WPP is normalized in this example.



Different Coordinate Spaces

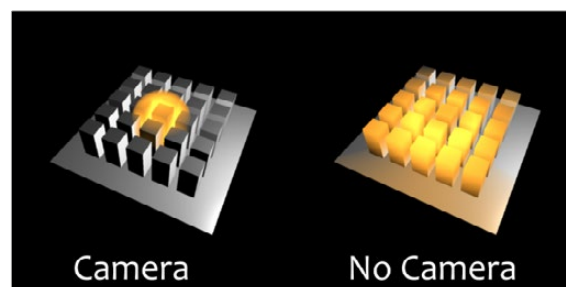
Rendering WPPs can occur in different Coordinate Spaces. These include World Space, Eye Space, and Object Space. The image below depicts how those different spaces look, although the nodes in Fusion require the WPP rendering to occur in World Space.



The Scene Input

The nodes offer a Scene input, which can either be a 3D camera or a 3D scene containing a camera. While the camera is vital for the Z to World node, Volume Mask and Volume Fog can generate their output without any camera attached or with the camera position set to 0/0/0.

However, connecting a camera that lines up with the original camera the WPP has been rendered from, or setting the camera's position manually, dramatically improves the accuracy and look of the resulting fog or mask.



The “Invisible Sphere”

The example scene shown so far has an empty background, meaning there is nothing in the scene apart from the ground plane and the cubes.

If applying fog to a scene like that, which is larger than the ground plane, the result will look similar to the “w/o Sphere” example shown below because, with no WPP information outside the ground plane, the resulting value is 0/0/0, and the fog fills that area as well.

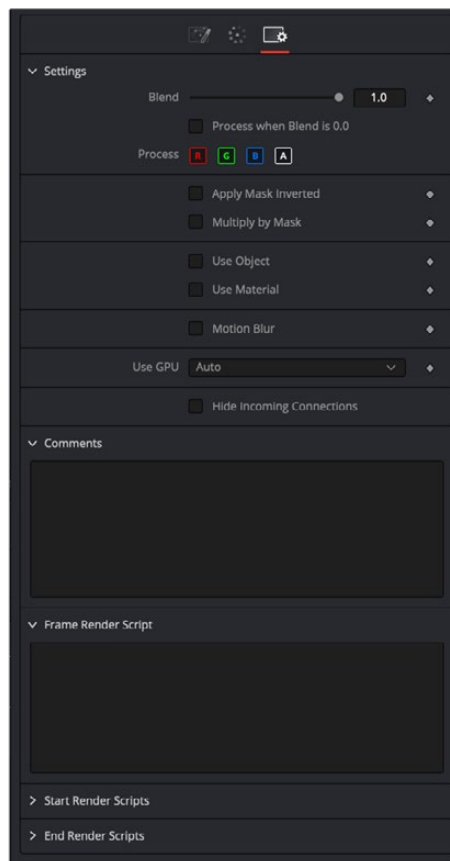
To get around that, you can add an invisible bounding sphere to your scene to create “dummy” WPP values to help the Fog node to create the correct volume as shown in the “with Sphere” example below.



The Common Controls

Nodes that handle Position operations share several identical controls in the Inspector. This section describes controls that are common among Position nodes.

Inspector



Position nodes Common Settings tab

Settings Tab

The Settings tab in the Inspector can be found on every tool in the Position category. The controls are consistent and work the same way for each tool.

Blend

The Blend control is used to blend between the tool's original image input and the tool's final modified output image. When the blend value is 0.0, the outgoing image is identical to the incoming image. Normally, this will cause the tool to skip processing entirely, copying the input straight to the output.

Process When Blend Is 0.0

The tool is processed even when the input value is zero. This can be useful if the node is scripted to trigger a task, but the node's value is set to 0.0.

Red/Green/Blue/Alpha Channel Selector

These four buttons are used to limit the effect of the tool to specified color channels. This filter is often applied after the tool has been processed.

For example, if the Red button on a Blur tool is deselected, the blur will first be applied to the image, and then the red channel from the original input will be copied back over the red channel of the result.

There are some exceptions, such as tools for which deselecting these channels causes the tool to skip processing that channel entirely. Tools that do this will generally possess a set of identical RGBA buttons on the Controls tab in the tool. In this case, the buttons in the Settings and the Controls tabs are identical.

Apply Mask Inverted

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

Multiply by Mask

Selecting this option will cause the RGB values of the masked image to be multiplied by the mask channel's values. This will cause all pixels of the image not included in the mask (i.e., set to 0) to become black/transparent.

Use Object/Use Material (Checkboxes)

Some 3D software can render to file formats that support additional channels. Notably, the EXR file format supports Object and Material ID channels, which can be used as a mask for the effect. These checkboxes determine whether the channels will be used if present. The specific Material ID or Object ID affected is chosen using the next set of controls.

Correct Edges

This checkbox appears only when the Use Object or Use Material checkboxes are selected. It toggles the method used to deal with overlapping edges of objects in a multi-object image. When enabled, the Coverage and Background Color channels are used to separate and improve the effect around the edge of the object. If this option is disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

For more information on the Coverage and Background Color channels, see *Chapter 78, "Understanding Image Channels,"* in the DaVinci Resolve Reference Manual, or Chapter 18 in the Fusion Reference Manual

Object ID/Material ID (Sliders)

Use these sliders to select which ID will be used to create a mask from the object or material channels of an image. Use the Sample button in the same way as the Color Picker: to grab IDs from the image displayed in the viewer. The image or sequence must have been rendered from a 3D software package with those channels included.

Motion Blur

- **Motion Blur:** This toggles the rendering of Motion Blur on the tool. When this control is toggled on, the tool's predicted motion is used to produce the motion blur caused by the virtual camera's shutter. When the control is toggled off, no motion blur is created.
- **Quality:** Quality determines the number of samples used to create the blur. A quality setting of 2 will cause Fusion to create two samples to either side of an object's actual motion. Larger values produce smoother results but increase the render time.
- **Shutter Angle:** Shutter Angle controls the angle of the virtual shutter used to produce the motion blur effect. Larger angles create more blur but increase the render times. A value of 360 is the equivalent of having the shutter open for one full frame exposure. Higher values are possible and can be used to create interesting effects.
- **Center Bias:** Center Bias modifies the position of the center of the motion blur. This allows for the creation of motion trail effects.
- **Sample Spread:** Adjusting this control modifies the weighting given to each sample. This affects the brightness of the samples.

Use GPU

The Use GPU menu has three settings. Setting the menu to Disable turns off GPU hardware-accelerated rendering. Enabled uses the GPU hardware for rendering the node. Auto uses a capable GPU if one is available and falls back to software rendering when a capable GPU is not available.

Hide Incoming Connections

Enabling this checkbox can hide connection lines from incoming nodes, making a node tree appear cleaner and easier to read. When enabled, empty fields for each input on a node will be displayed in the Inspector. Dragging a connected node from the node tree into the field will hide that incoming connection line as long as the node is not selected in the node tree. When the node is selected in the node tree, the line will reappear.

Comments

The Comments field is used to add notes to a tool. Click in the empty field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.